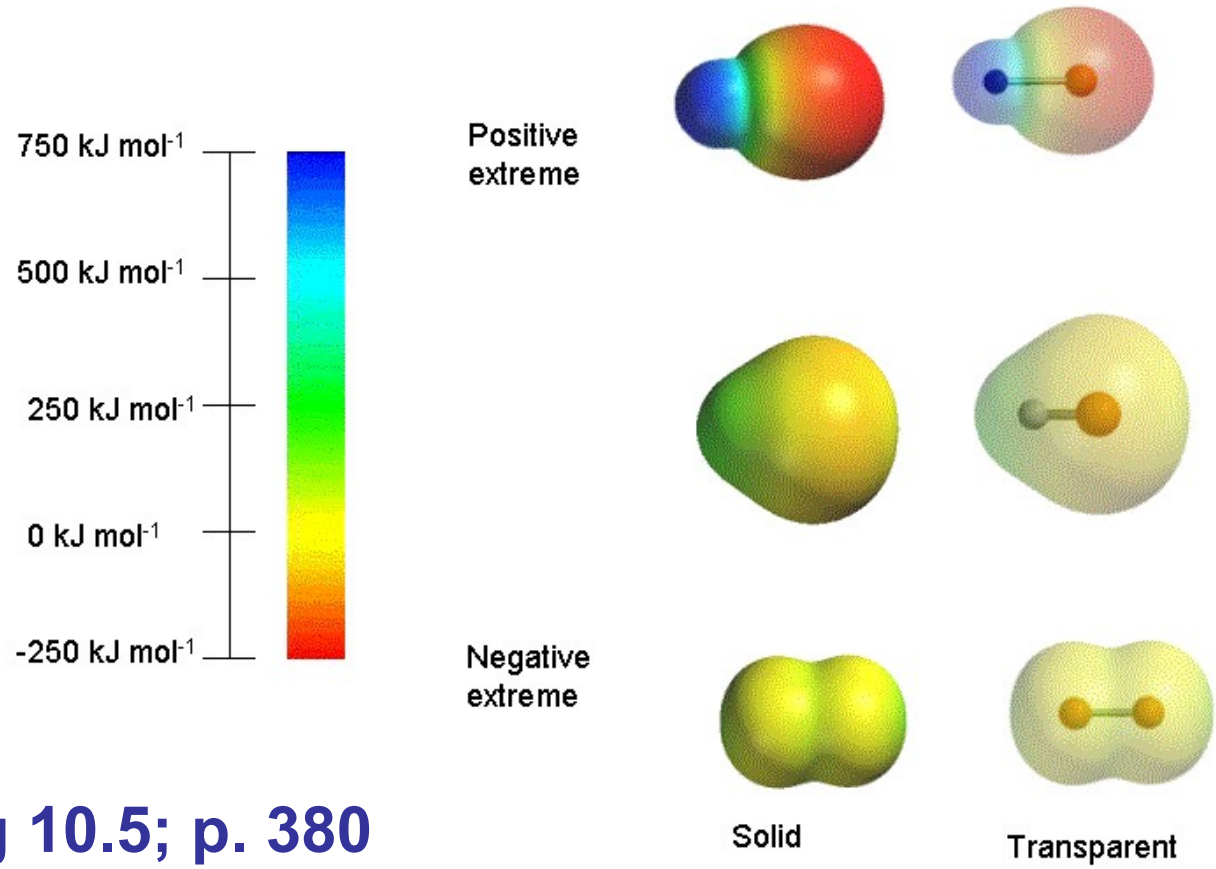
Chapter 3
Polar Bonds and polar Molecules

Polar Covalent Bonds

- Homonuclear molecules (*eg* F_2, H_2) are nonpolar (electrons shared equally).
- In heteronuclear molecules (*eg* HCl, HF) the bonds are polar (electrons not shared equally).



NaCl (ionic)

HCl (polar covalent)

Cl_2 (covalent)

Fig 10.5; p. 380

Electronegativity

- In polar bonds, electrons are attracted to the atom with a greater electronegativity.
- Electronegativity (EN) is the ability of an atom in a molecule to attract shared electrons to itself.
- Why not use electron affinity?
  - EA is an atomic property, EN is a molecular property.
- A widely used scale is the Pauling electronegativity scale.

Increasing Electronegativity

Increasing Electronegativity

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
H 2.1																
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5
Cs 0.8	Ba 0.9	La* 1.1	Hf 1.3	Ta 1.5	W 2.4	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.8	Bi 1.9	Po 2.0	At 2.2
Fr 0.7	Ra 0.9	Ac† 1.1														

below 1.0
1.0–1.4
1.5–1.9
2.0–2.4
2.5–2.9
3.0–4.0

*Lanthanides: 1.1–1.3
†Actinides: 1.3–1.5

Fig 10-6, p. 382

Electronegativity

- The bigger the electronegativity difference (ΔEN), the more polar the bond and the closer it will be to ionic rather than polar covalent.

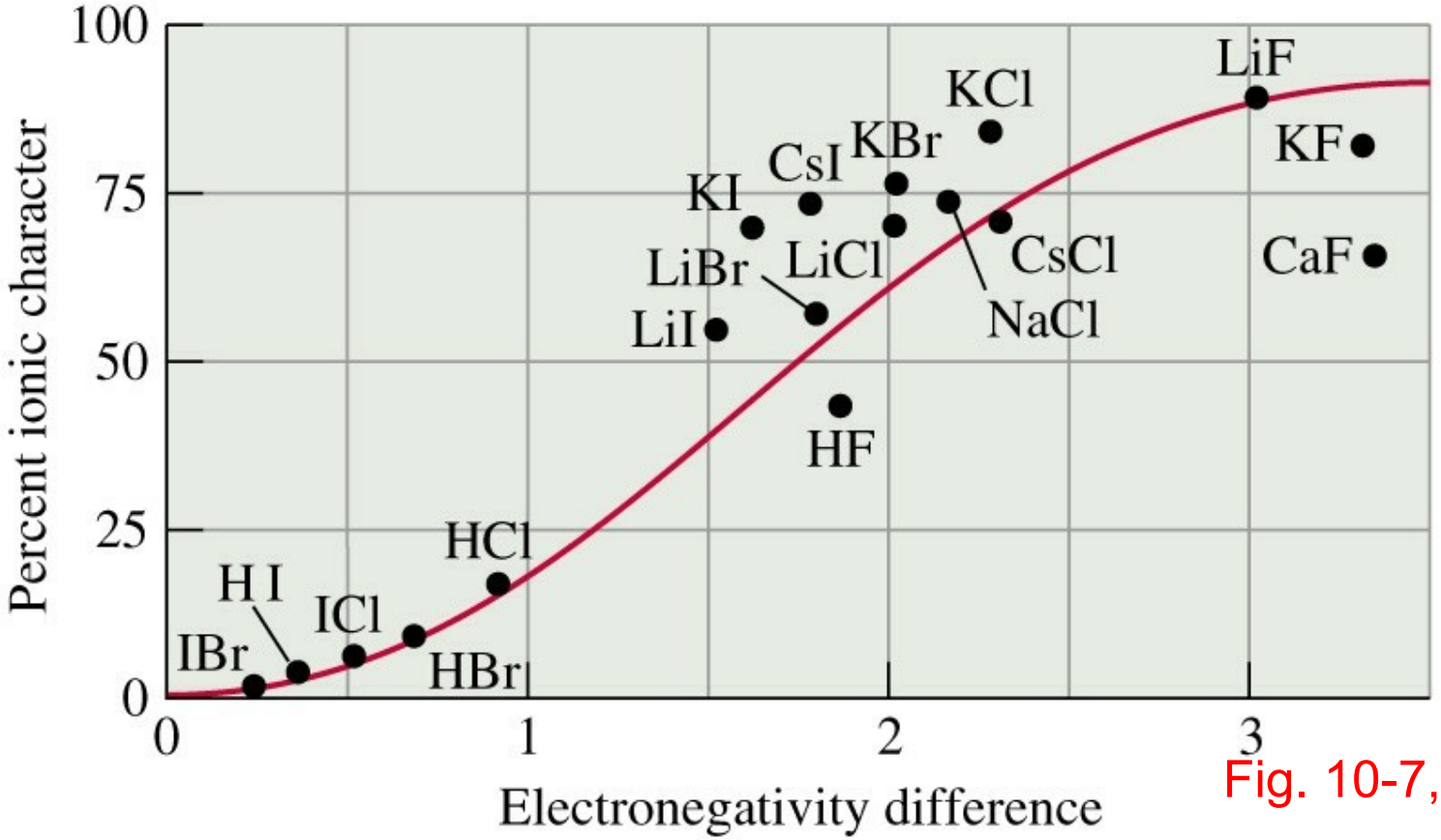
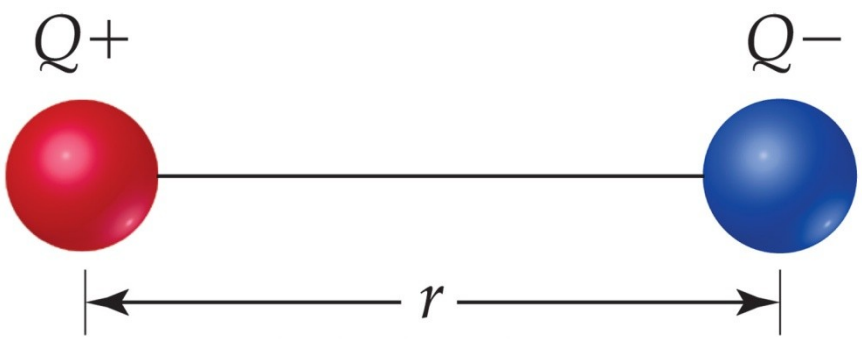


Fig. 10-7, pg 382

Dipole Moments

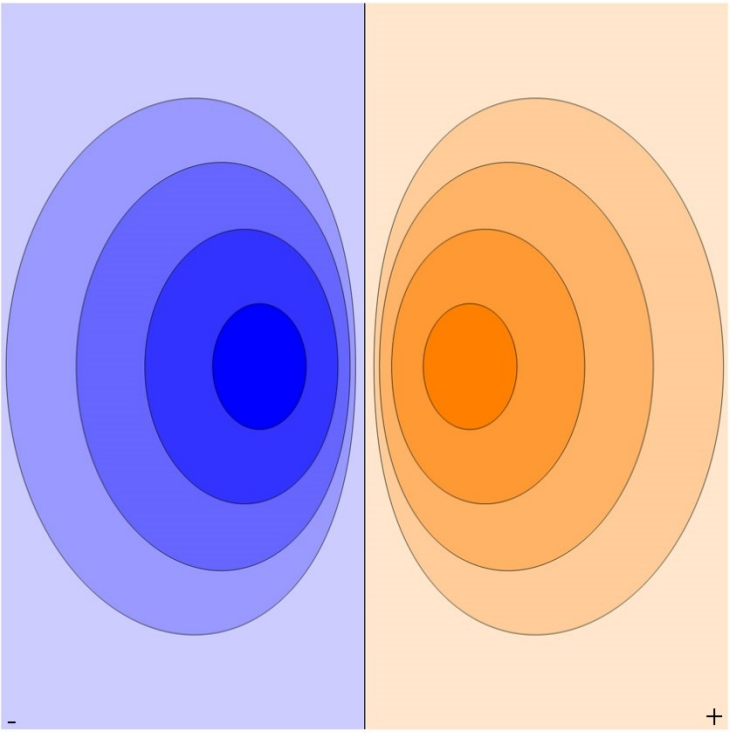
- We can actually measure how polar a bond is! When two atoms share electrons unequally, a bond dipole results.



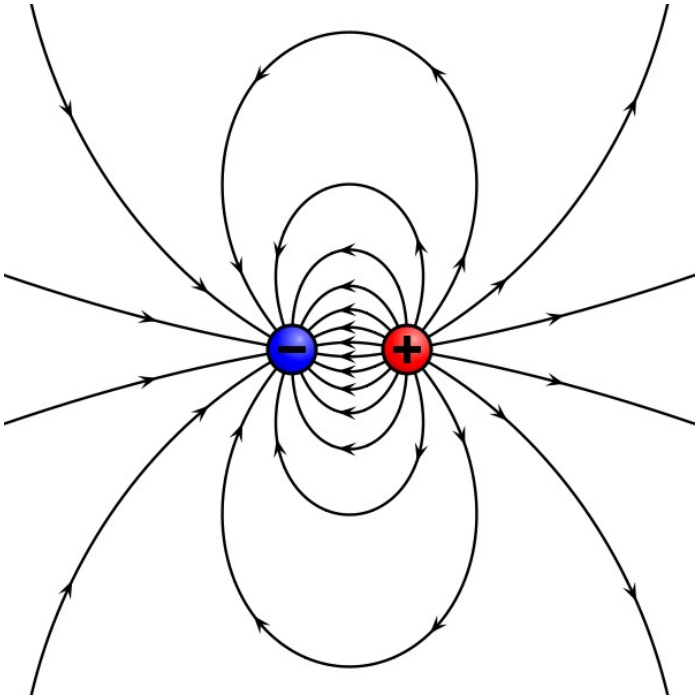
- The dipole moment, μ, produced by two equal but opposite charges separated by a distance, r, is calculated:

$$\mu = Qr$$

- It is measured in debyes (D).



Contour plot of the electrostatic potential of a horizontally oriented electrical dipole of finite size. Strong colors indicate highest and lowest potential (where the opposing charges of the dipole are located).



Electric field lines of two opposing charges separated by a finite distance

Source of pictures *Wikipedia*

Polar Covalent Bonds

The greater the difference in electronegativity, the more polar is the bond.

Compound	Bond Length (Å)	Electronegativity Difference	Dipole Moment (D)
HF	0.92	1.9	1.82
HCl	1.27	0.9	1.08
HBr	1.41	0.7	0.82
HI	1.61	0.4	0.44

Table 7.3

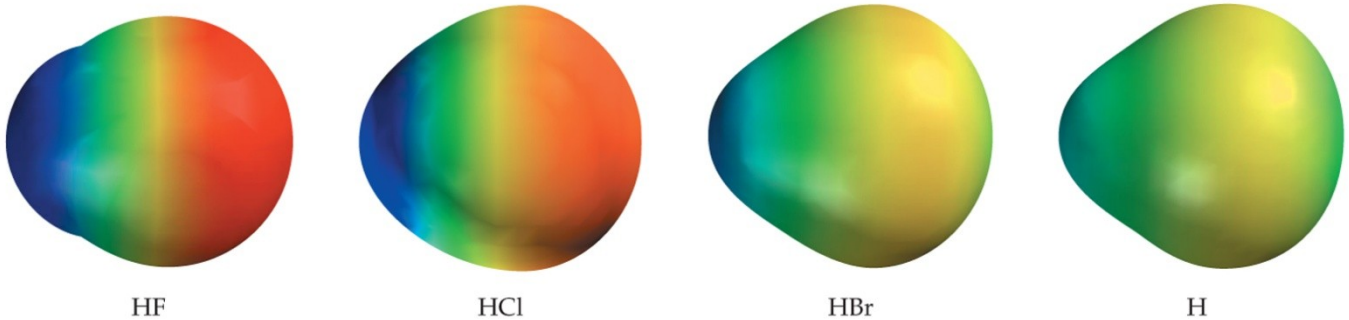


Figure 7.8

Polar molecules

In polar molecules fractional (or partial) positive and negative charges arise are indicated as δ+ and δ-

Consider, **HCl**

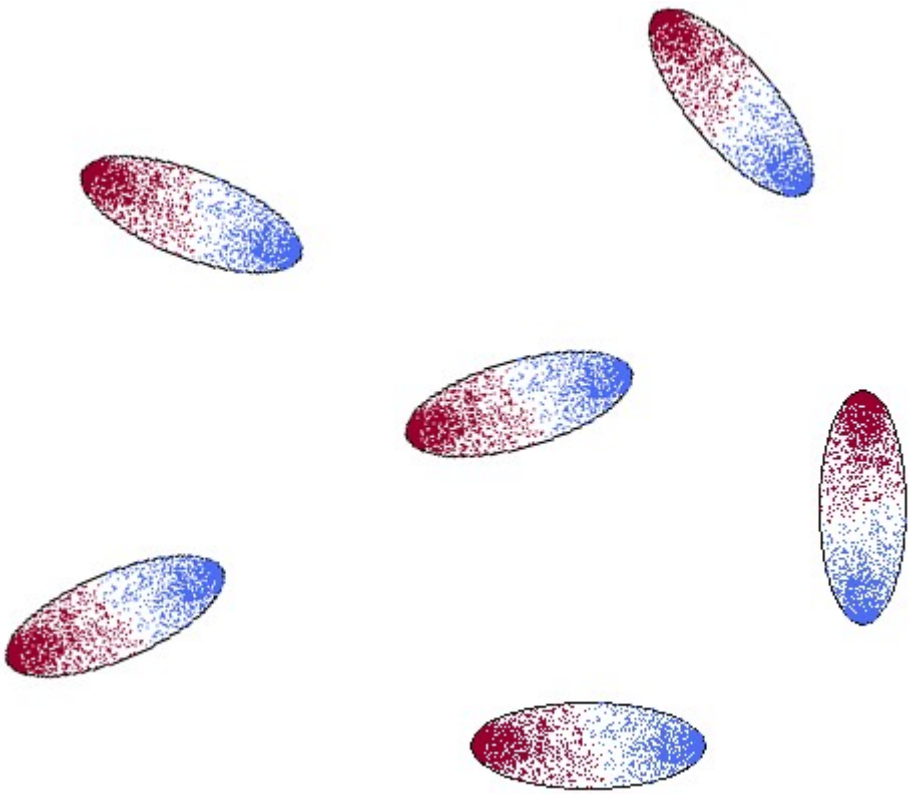
Dipole moment,

μ =fractional charge X distance between nuclei

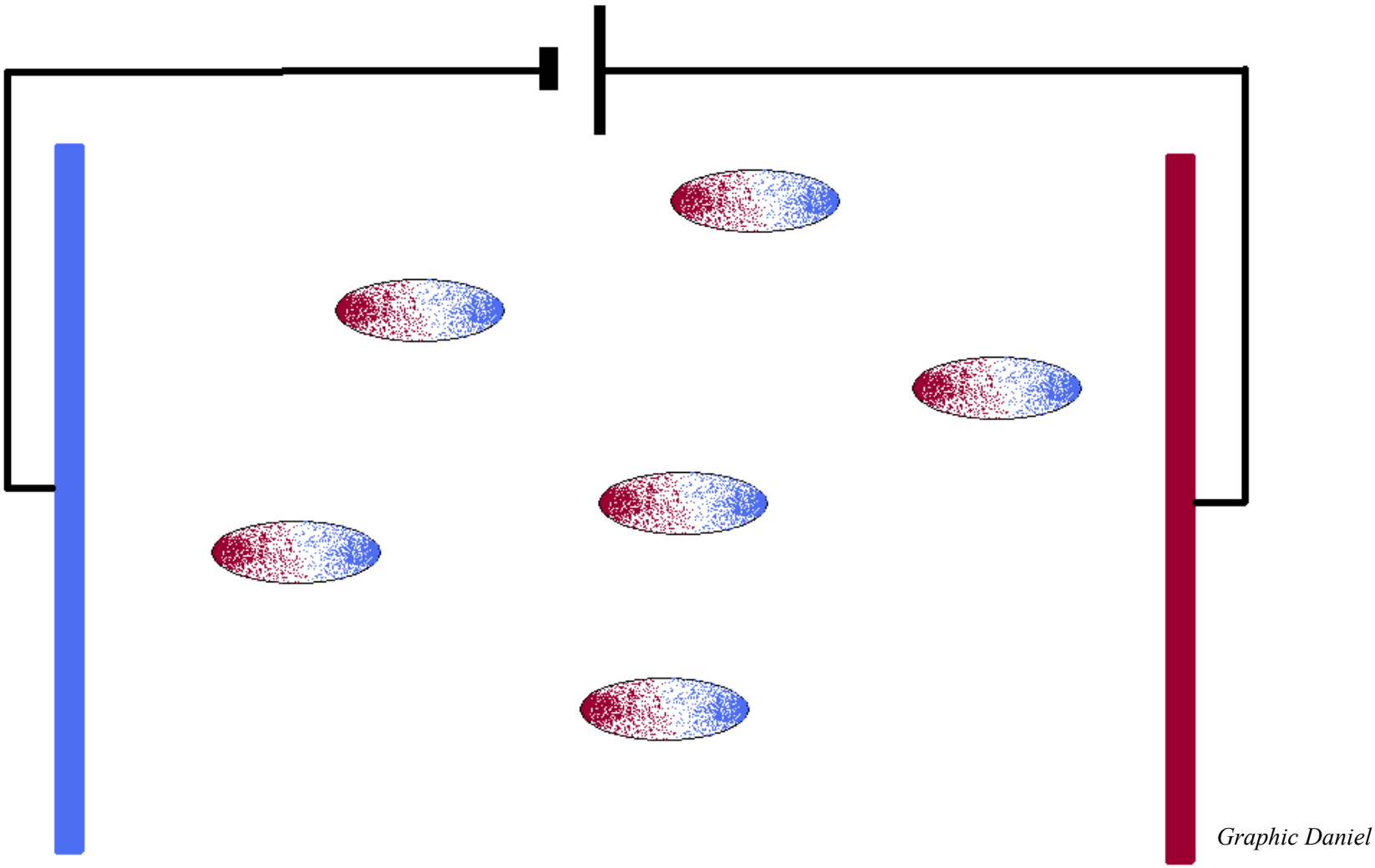
Calculate the magnitude of δ+ and δ- (e.g. the fractional charge) in a HCl molecule as well as the % ionic character of the HCl bond. The bond distance in HCl is 126 pm and the dipole moment was measured to be 1.08 Debye (D)
(1 D = 3.34×10^{-30} Cm and the elemental charge is 1.602×10^{-19} C)

How do we measure dipole moments?

Polar HCl molecules in the absence of an external electric fiel



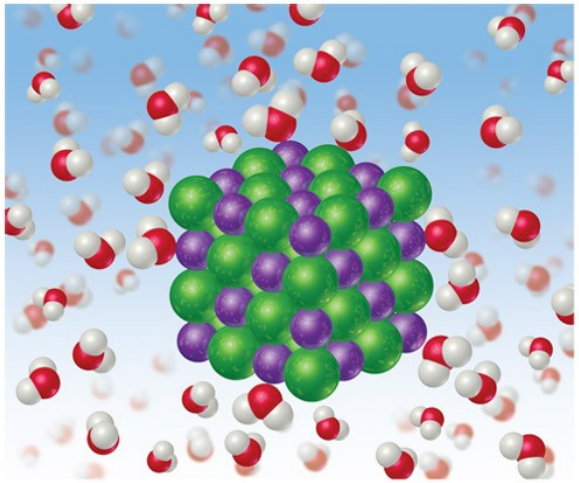
Graphic Daniel

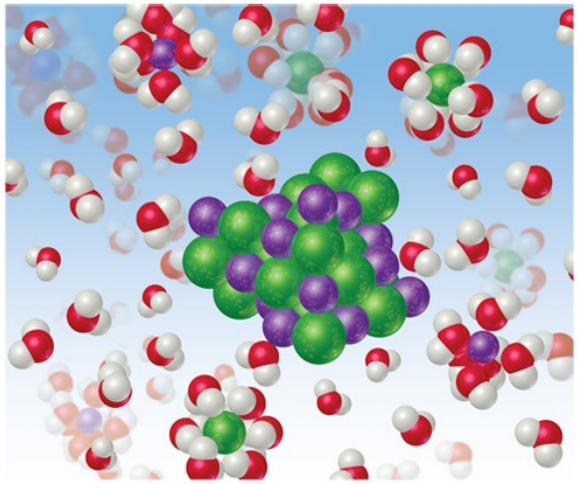


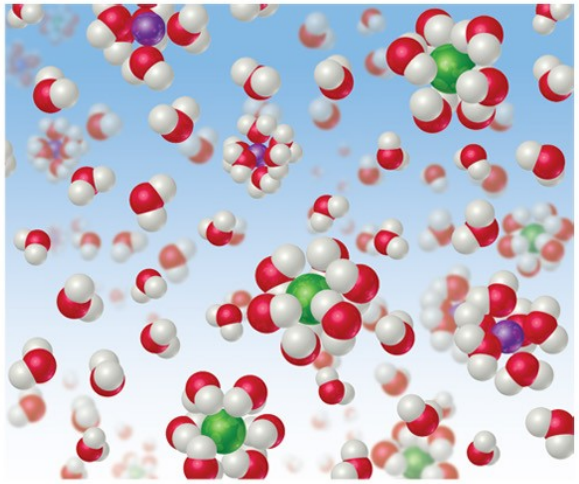
hydrogen chloride molecules orient themselves in an electric field. **As a result, the field is weakend**.

So what about a “molecular capacitor” such as **Na^+ Cl^-** separated by vacuum and introduced to H_2O?

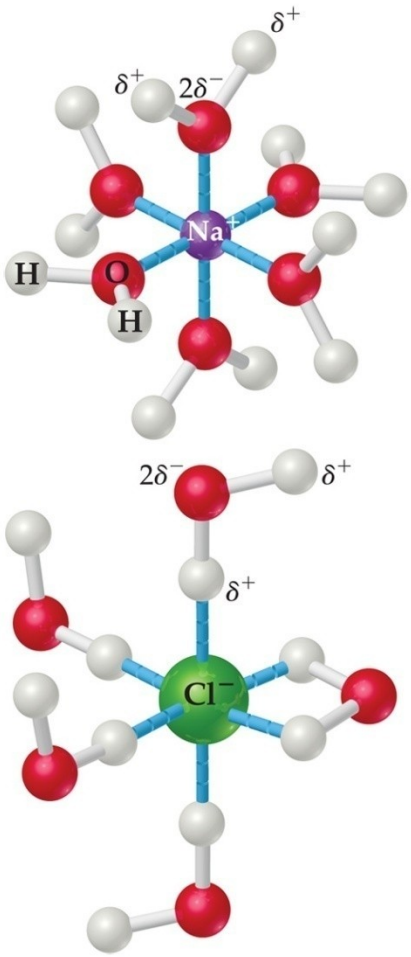
Polar water molecules weaken the electric field between sodium and chloride ions by almost two orders of magnitude. NaCl (mp 801 °C) spontaneously dissolves in water at room temperature.







The Solution Process



In addition ion-dipole interactions cause solvation of the ions which essentially shields ion of opposite charge from electrostatic attraction.

Brown LeMay

Give it some thought!

- Why do some ionic compounds dissolve in water while others do not (for example MgO)
- Explain why NaCl dissolves in water but not in gasoline.

All heteronuclear diatomic molecules must be polar, but what about larger molecules?

Molecular Shape and Molecular Polarity

If a molecule possesses polar bonds it does not mean the molecule *as a whole* will be polar.

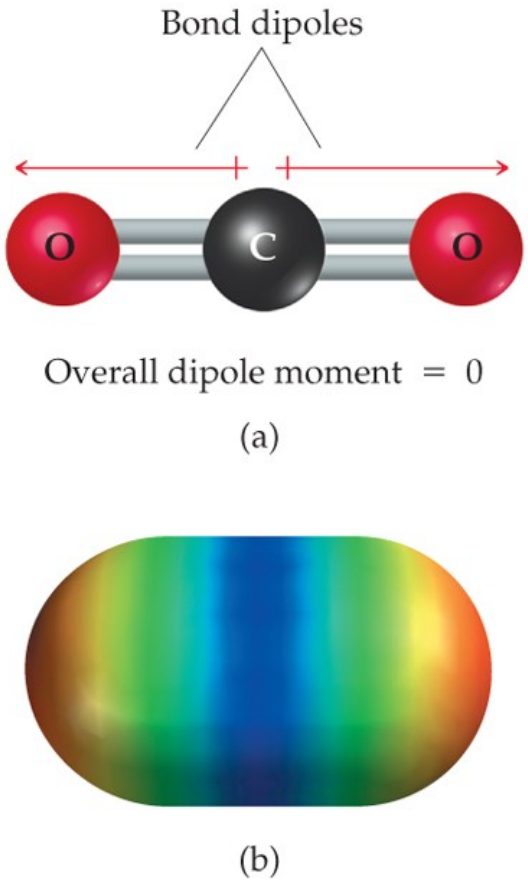
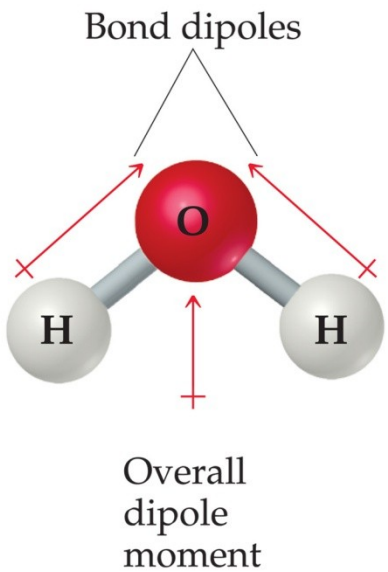


Figure 8.11

Molecular Shape and Molecular Polarity



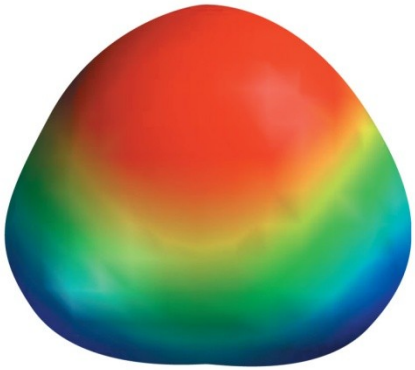


Figure 8.12

By "adding" the individual bond dipoles, one can determine the overall dipole moment for the molecule.

Molecules Containing Polar Bonds

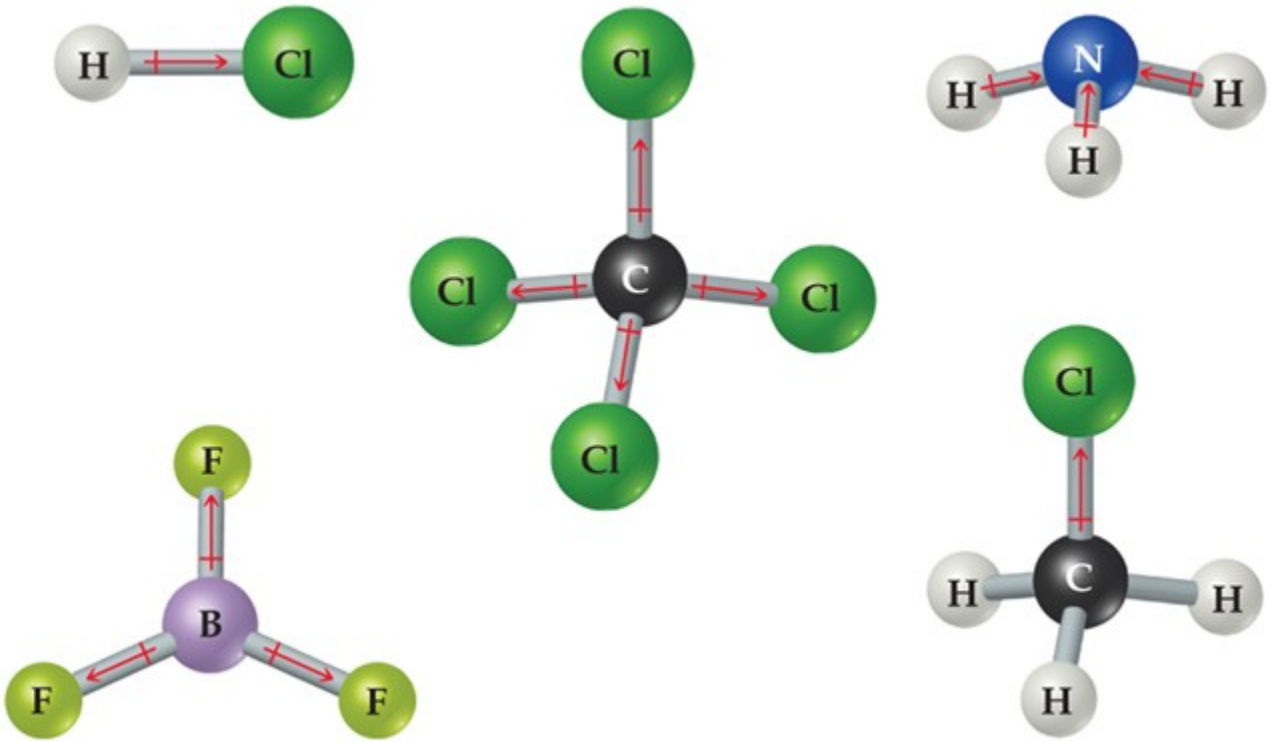
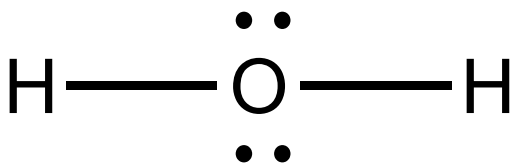


Figure 8.13

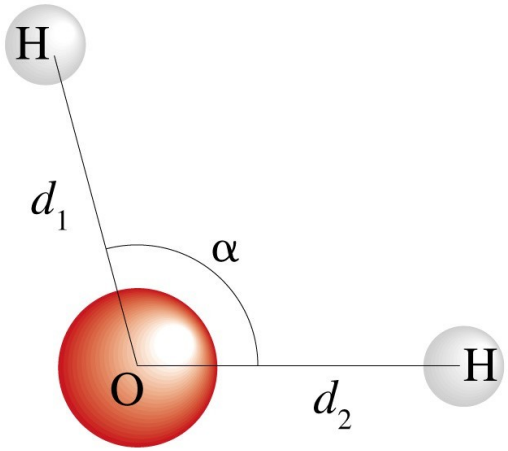
In short:
To determine if a molecule or ion is polar or not, we need to know its molecular shape!

The Shapes of Molecules

Lewis structures are useful but don't allow prediction of the shape of a molecule.



water



Can use a simple theory based on electron repulsion to predict structure (for non-transition elements).

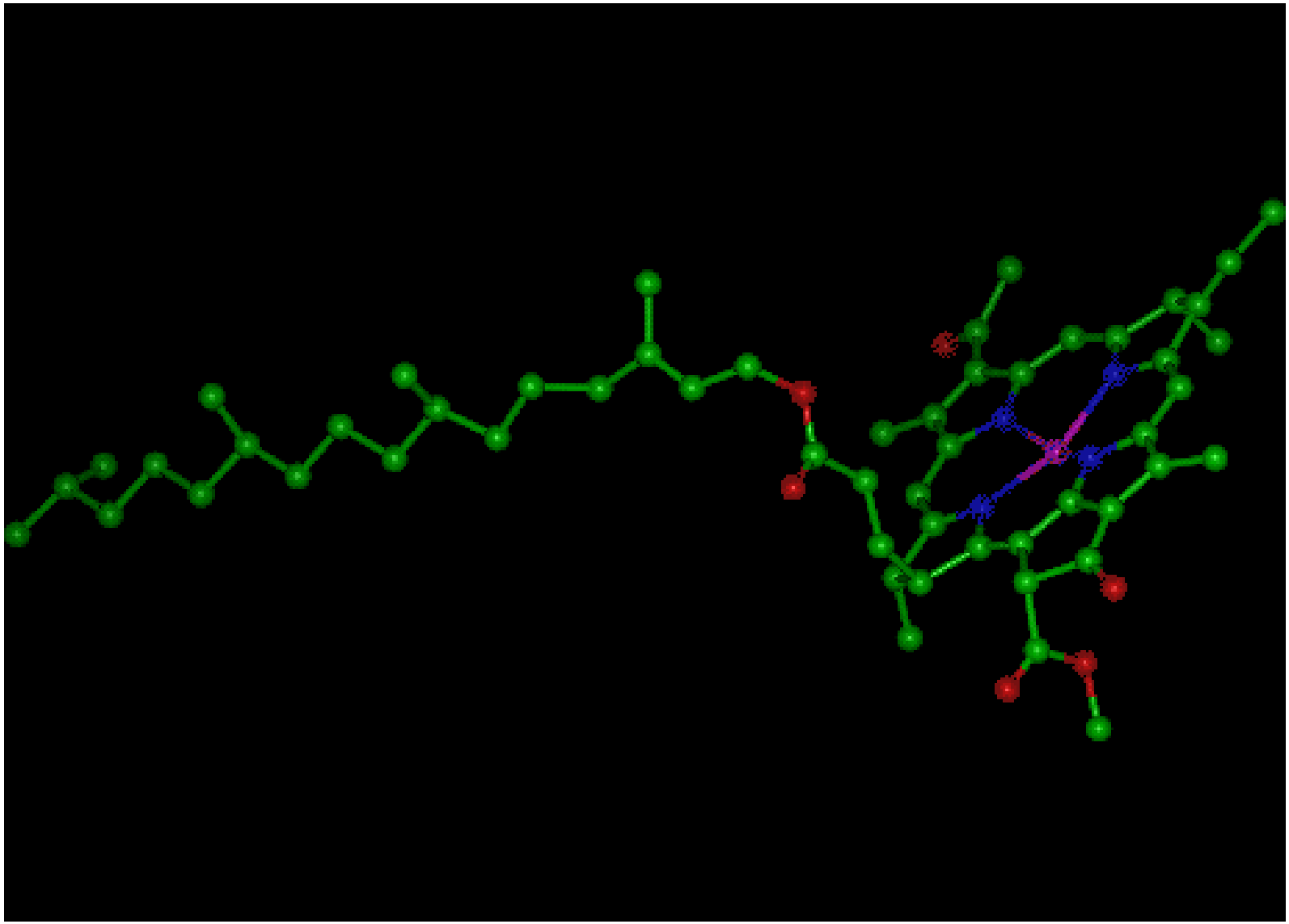
VSEPR Theory

Valence-Shell Electron-Pair Repulsion Theory

- Electron pairs repel each other whether they are in chemical bonds(X) or lone pairs (E).

- Valence electron pairs are oriented to be as far apart as possible to minimize repulsions ==> LOWEST ENERGY!

Chlorophyll



Electron Group Geometry (EGG)

You must know how to draw these!

EGG = Lone Pairs (E) + Bond Pairs(X)

Linear	**(E + X = 2)**
Trigonal Planar	**(E + X = 3)**
Tetrahedral	**(E + X = 4)**
Trigonal bipyramidal	**(E + X = 5)**
Octahedral	**(E + X = 6)**
Pentagonal bipyramidal	(E + X = 7)

Shapes of Molecules

- All molecular shapes are based on the EGG.
- To determine the EGG:
  1. Draw a Lewis Structure
  2. Determine # Bond Pairs (X) - ***multiple bonds count as 1 only.***
  3. Determine # Lone Pairs (E)
  4. EGG = X + E

EGG is also known as "Parent Shape"

Shapes of Molecules

- Once the EGG is known, lone pairs (E) are placed in positions to minimize E-E and X-E repulsions. E-E repulsions are larger than X-E repulsions.
- The Molecular Geometry (aka SHAPE) is determined by the position of the bond pairs (X).

Examples (VSEPR)

(i) Determine the molecular geometry (shape) of each.
(ii) Draw a perspective diagram.
(iii) Polar or non-polar?

(a) HCN

(b) $[NO_2]^-$

(c) CH_4

(d) NH_3

(e) IF_3

(f) BrF_5

(g) H_2O

(h) I_3^-

For Homework: SF_4, PF_5, SF_6, XeF_4, $[SiF_6]^{2-}$